0:01

Let's say that this is you.

0:02

You're enjoying a nice sunny day,

0:05

and you decided to take a nice long, deep breath of air.

0:09

And of course, when I say air, the part

0:11

that you probably care the most about

0:14

is just the oxygen part of that air.

0:16

That's the part that we, as humans, need to survive.

0:20

So you take a deep breath, and let's say

0:23

that you take it through your mouth.

0:24

You take a deep breath through your mouth.

0:26

And then let's say you take one more deep breath,

0:29

a second deep breath.

0:30

And you take that one through your nose.

0:31

Now you might think, well, these are two totally different ways

0:35

of getting in air.

0:36

That's certainly how it looks when

0:38

you look at a mouth and a nose.

0:40

It doesn't look like they have much in common.

0:42

But the truth is that actually.

0:44

If you follow the air, it almost follows an identical path.

0:49

So the air is going to go into the back of the throat,

0:53

really regardless of how you took it in.

0:56

So here we have air coming in from the nose.

0:59

And here you have air coming in from the mouth.

1:01

And they meet up in the back of the throat.

1:04

And then they go down, down, down.

1:06

They go towards this thing that we call the Adam's apple.

1:10

I'm going to bring up a little bit of the canvas

1:12

so you can see it more easily.

1:15

But basically, you see this Adam's apple right here.

1:19

And actually you can go ahead and take

1:20

a feel of your own Adam's apple.

1:23

It's a pretty cool structure in the middle of your throat,

1:27

and everybody has it.

1:28

That's the first thing I want to tell

1:30

you is that everybody has it, not just men.

1:33

Women have it too.

1:34

And the reason it's called an Adam's apple,

1:37

it's called an Adam's apple because Adam is generally

1:41

a boy's name.

1:42

And so it's to remind us that usually men or boys have

1:47

larger Adam's apples than girls.

1:50

And if you're trying to find it, I just also want to point out

1:53

there's this notch here.

1:54

And if you can feel the notch with your fingers,

1:57

that gives you a nice clue as to where it's located.

2:01

But this is it.

2:02

This is the Adam's apple.

2:03

And what it does is it helps you control your voice.

2:06

And actually, another name for the Adam's apple, sometimes

2:10

people call it the voice box.

2:12

The voice box.

2:14

And of course, air is passing through the voice box,

2:17

and it's kind of the entryway into the trachea.

2:21

And so it actually allows me to (FALSETTO)

2:24

make my voice really high, (DEEP BARITONE)

2:26

or make my voice very low, depending

2:29

on how you change the muscles around in that Adam's apple.

2:33

So that's actually the kind of first cool thing

2:35

I want to point out to you is that you can actually

2:38

control your voice.

2:39

I'm sure you knew this already, but what you're using

2:42

is your Adam's apple, or your voice box.

2:45

Now air keeps going.

2:46

Air is just going to keep making its journey down--

2:48

and specifically, of course, the part of air

2:51

I said we care about is the oxygen.

2:54

It's going to keep making its journey down

2:56

into the lung area.

2:58

So this is now the lung area.

2:59

It's going down the trachea, and it goes into the two lungs--

3:03

the right and left lung.

3:05

This is the left lung.

3:07

I'm going to put L for left.

3:08

And this is the right lung.

3:09

I'll put R for right.

3:11

And immediately you're thinking, well, wait a second,

3:14

aren't they switched?

3:15

I want you to remember that this is

3:16

from the perspective of the person who owns the lungs,

3:20

so that's why I'm putting left where I put it

3:22

and right where I put it.

3:23

Now we should probably go ahead and start

3:25

labeling some of this.

3:26

You can see that the lungs actually don't look identical.

3:30

They look slightly different.

3:32

For example, this one has three lobes.

3:35

The right side has three lobes.

3:36

We call it the upper lobe, middle lobe and lower lobe.

3:40

And the left one only has two lobes.

3:43

So that's the first kind of big difference.

3:46

And the other difference is that you actually

3:49

have this thing in the middle that we call a cardiac notch.

3:53

This thing right here, this is called the cardiac notch.

3:57

And the reason we call it that is

3:59

that it's a little spot that gets

4:02

formed because the heart is literally

4:05

kind of peeking out here.

4:08

And as a result, it kind of makes a notch

4:10

in the lungs when it develops.

4:12

So the heart takes up a little bit of space here.

4:14

This is our heart.

4:16

And as a result, it makes that notch.

4:20

So this is our heart space here.

4:23

So on the other side, you've got,

4:24

of course, your two lobes-- your upper and lower lobe.

4:27

And these are two clues, so if you ever see a lung just kind

4:30

of sitting by itself and you want to figure out

4:32

whether it's the left lung or the right lung,

4:35

you can look for the number of lobes,

4:38

or you can look for that cardiac notch.

4:41

Now around these lungs, you've got ribs.

4:45

So you've got ribs here, and between the ribs,

4:48

you've got rib muscles.

4:49

And they are, of course, on both sides.

4:52

And below the lungs and below the heart,

4:56

you've got a big muscle.

4:59

Actually it's going to come through here.

5:02

I'm going to just kind of go through the word heart.

5:06

And it basically becomes the floor.

5:08

So the heart and the two lungs, they

5:10

sit on this floor made up of this muscle.

5:14

And this muscle is the diaphragm muscle.

5:18

So this diaphragm muscle makes up the floor.

5:22

The ribs make up the walls.

5:25

So what do we have?

5:27

We have basically a room.

5:28

We have a giant room with walls and a floor.

5:32

And this entire room we actually call the thorax.

5:36

So within this room then, you have your two lungs

5:39

and your heart.

5:41

So far, so good.

5:42

But I haven't done a very nice job of actually showing you

5:45

where the air goes.

5:46

I've just kind of pointed that it goes through the two lungs,

5:48

but you don't actually get to see where it goes after that.

5:51

So I'm going to erase a lot of this.

5:54

I'm going to reveal to you what it

5:56

would look like if you could slip on some X-ray glasses

6:00

and look at your two lungs.

6:01

This is kind of what it would look like.

6:03

You've got all this interesting architecture.

6:08

And the easiest way to think about this, probably

6:11

the simplest way to think about this, is to imagine a tree.

6:14

So imagine a tree, and that tree has been flipped upside down.

6:19

So you've got all these branches off that tree.

6:22

And they're branching and branching.

6:25

And if you flip this tree upside down,

6:27

you start seeing that it looks a lot like what

6:30

we have in our lungs.

6:31

Our lungs basically look like a flipped upside down tree.

6:36

And we even call it that.

6:37

We even call this entire structure,

6:39

we call it a bronchial tree.

6:42

So when you look at the lungs and they

6:45

look kind of messy or complicated,

6:47

just think of them as an upside down bronchial tree, and all

6:51

of a sudden it'll look much simpler.

6:54

Basically in the middle, you've got this nice trunk.

6:57

Right?

6:57

This is our trunk.

6:58

And then it starts kind of branching from there.

7:01

So air goes down this main trunk, this trachea, and then

7:04

it kind of starts splitting up.

7:06

And each of these colored regions-- the green region

7:10

or the purple region-- serves a different lobe.

7:13

So this green region serves the lower lobe down here.

7:17

The purple serves the upper lobe.

7:19

And on this side, you've got an upper, a middle, and a lower

7:22

lobe.

7:23

Now I know it looks a little bit strange, because you've

7:26

got some green branches in what should be the middle lobe,

7:29

like right here.

7:30

You've got some orange branches in what

7:32

looks like the upper lobe, like right there.

7:34

But what you have to remember-- and this

7:36

is kind of tricky to do and just try

7:38

to play with it in your head-- is

7:40

that what you have is basically a three-dimensional lung.

7:45

So you have to imagine that we are only

7:47

looking at it from the front side.

7:49

But, of course, that middle lobe does go back.

7:52

And if it went back, then you'd make perfect sense

7:55

of why the orange branches are where they're at.

7:58

Now let me continue the air journey,

8:00

because I want to make sure we finish it off.

8:02

So let's say we take a little branch like this, we expand it.

8:05

We keep zooming into it, zooming into it,

8:08

zooming into until it's microscopic.

8:10

You can't see it with your eyes anymore,

8:12

but you could see it under a microscope.

8:15

It would look like this.

8:16

It would basically, under a microscope,

8:18

look like a bunch of little sacs like this.

8:22

And these sacs, we call these alveoli.

8:25

Alveoli.

8:27

And the air actually kind of runs into the alveoli.

8:31

It has a dead end, and then it comes back around.

8:34

And then you breathe it out.

8:35

So that's how breathing works.

8:37

The air goes all the way in through your mouth, down

8:39

to the alveoli, takes a U-turn, and then goes back out.

8:42

But before it does that, before it leaves,

8:45

very close to the alveoli is blood.

8:49

And let's say blood is coming this way and going that way.

8:53

And what will happen is that actually into the blood

8:58

will go oxygen.

9:00

Oxygen will actually go into the blood.

9:02

And out of the blood will be waste.

9:05

So you'll have some carbon dioxide waste

9:07

that your cells have been making.

9:09

And that waste actually then gets

9:12

thrown back into the alveoli.

9:15

So now you can see how oxygen gets from the outside world,

9:18

gets breathed in through the lungs when you inhale,

9:21

gets down into the alveoli, exchanges with the blood,

9:25

and then you exhale and let all that carbon dioxide out.